Reg. No. | | | | | | | | | | | [12EE752]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER–2015

APPLIED EMBEDDED SYSTEM DESIGN

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions and all questions carry equal marks.

1.	**a.**	List the hardware units that must be present in the embedded systems.	**9**
	b.	Explain the exemplary applications of type of embedded system.	**5**
2.	**a.**	Explain the parallel port devices.	**7**
	b.	Explain the types of UART.	**4**
	c.	What is watch dog timer? Explain its operation.	**3**
3.	**a.**	Explain the interrupt handling mechanism.	**7**
	b.	With neat diagram explain the operation of DMA.	**7**
4.	**a.**	Explain basic architecture of ARM processor.	**7**
	b.	Explain the current program status register.	**4**
	c	Mention the banked registers available in ARM architecture.	**3**
5.	**a.**	What are the advantages of high level language codling?	**7**
	b.	Explain following C program elements i) Header files ii) Pre processive elements.	**7**
6.	**a.**	Explain inter process Communication Mechanism.	**7**
	b.	Explain shared memory communication and message- Based Communication.	**7**
7.	**a.**	Explain the goals of operating system services.	**7**
	b.	Explain the scheduler in which RTOS insert into the list and the ready task for sequential execution in a co-operative round robin model.	**7**
8.	**a.**	Explain the RTOS programming tool Micro C/OS-II.	**7**
	b.	Explain the features of V_x works.	**7**

SCHOOL OF ENGINEERING & TECHNOLOGY
LIBRARY
DATE
JAIN UNIVERSITY-BANGALORE